Homework 4
FIN2020 Fall 2023

Due at **Midnight 11:59pm Wednesday, Nov. 01 2022 Beijing Time**

Problem 1

Consider a portfolio allocation problem that is a special case of those we studied in class. An investor has initial wealth $Y_0 = 100$. The investor allocates the amount a to stocks, which provide return $r_G = 0.3$ in good state that occurs with probability $1/2$ and return $r_B = 0.05$ in a bad state that occurs with probability $1/2$. The investor allocates the remaining $Y_0 - a$ to a risk-free bond, which provides the return $r_f = 0.1$ in both states. Assuming that the investor has vN-M expected utility, with Bernoulli utility function of the logarithmic form

$$u(Y) = \ln(Y).$$

Calculate the optimal amount a^* that the investor should allocate to stocks.

Problem 2

Consider an optimal saving problem under uncertainty in a two-period economy, $t = 0$ and $t = 1$. Assume the consumer with initial wealth Y_0 has Bernoulli utility of the logarithmic form, and the consumer is trying to maximize expected utility of total consumptions at both periods:

$$\ln(c_0) + \beta \mathrm{E}[\ln(\tilde{c}_1)],$$

where β is the time discount factor that measures how patient the consumer is.

Suppose in this economy, there exists only one saving device, a stock, which provides return r_G in good state that occurs with probability π and return r_B in a bad state that occurs with probability $1 - \pi$.

Then what is the optimal amount of saving in stock for this consumer? How does the optimal saving depend on π, and can we generalize this dependency property to any Bernoulli utility function?

Problem 3

Consider two risky assets with random payoffs $\tilde{z}_1$ and $\tilde{z}_2$:

$$\tilde{z}_1 = \begin{cases} 1 & \text{with probability } 0.25 \\ 7 & \text{with probability } 0.50 \\ 12 & \text{with probability } 0.25 \end{cases}, \quad \tilde{z}_2 = \begin{cases} 3 & \text{with probability } 0.33 \\ 5 & \text{with probability } 0.33 \\ 8 & \text{with probability } 0.34 \end{cases}.$$

Check that neither asset dominates the other on the basis of

(a) The Mean-Variance Dominance

(b) First Order Stochastic Dominance

(c) Second Order Stochastic Dominance